

SCI-MAP

Ordinary Mapmaking and Observation Coaddition

Engineering Conformance Specification

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1 Purpose and conformance target

This specification tells a reviewer how to decide whether one exact candidate conforms to SCI-MAP v0.2. It does not select a candidate, inspect an implementation, execute validation, or assert a result. Before work begins, the reviewer freezes:

- repository and immutable candidate identity;
- build profile, compiler/runtime identities, configuration, and enabled parallel modes;
- exact input fixtures and independent expected-value generators;
- numerical comparison policy and simultaneous statistical error policy;
- required product list, representation paths, log policy, and completion criteria; and
- the shared-authority digest, contract version, this document revision, owner-decision ledger revision, exact byte-identical PTC-to-MAP boundary digest, exact original-footprint-coordinate boundary, source-bound MAP profile versions and their six separate status axes, six product-role identities, coadd Registry record, owner r0.4 directive identity/digest, and the exact candidate paired `SCI-VAL_PROFILE_REGISTRY v0.1/r0.3-map-v0.2-r0.4-candidate-2026-09-08` and `SCI-VAL_SOURCE_BINDING_REGISTER v0.1/r0.3-map-v0.2-r0.4-candidate-2026-09-08` records. Candidate records are not treated as owner-accepted, canonically activated, or frozen merely because their bytes and hashes are complete.

Evidence from a different candidate, configuration, or science-authority digest is not silently pooled. Each rerun keeps its own identity and reason. Historical operational acceptance does not substitute for conformance evidence.

The complete ordered MAP lifecycle, its independent attempt outcome, and its record rules appear once in the following lifecycle table.

MAP stage	Exact meaning	Record rule
requested	Immutable accepted intent.	Exists only for an accepted request; it is never reconstructed from a later stage.
effective	Supported executable interpretation of that request.	Binds supported/disabled/unavailable interpretation without mutating intent.
observation_resolved	Every observation-specific parent, coefficient, profile, geometry, mapping, parameter, support rule, output role, and persistence decision is concrete or typed unavailable.	Supplies the exact value/state for pre-application validation.
applied	Exact resolved operation and parameter state was invoked or committed for one execution attempt.	Created only at the application event; retained with exact terminal failure when output realization fails.
realized	Actual immutable accumulators, scientific output rows, product values, role-qualified companion statuses, product identity, and provenance were produced.	Created only for an actual MAP product; a failure- evidence record is not product realization.
Attempt outcome	Exactly succeeded , failed , or not_produced .	Recorded independently of the reached lifecycle prefix and product realization.
Termination rule	Stop before application; post-application terminal failure; or applied valid policy with no support-authorized rows.	The first has no applied or realized record. The latter two retain the applied-attempt record and exact cause but no realized MAP product; the no-support case is not_produced , not failed .

The MAP plan, coefficient selection/application, projection, support policy, output/coadd plan, response/covariance request, product role, and **coverage_cut** use this same ordering. Each parameter stage binds its exact value and provenance; a completed realization contains all five stages, while a nonproducing attempt contains only stages actually reached.

The MAP lifecycle does not replace the independently named PTC/VAL handoff axes **requested/not_requested**, **applicable/inapplicable/applicability_unknown**, **eligible/ineligible/decision_unavailable**, and **realized/incomplete/failed/not_produced**. It also does not replace the ECS requirement-result vocabulary or evidence- artifact realization state. In particular, a MAP parameter's resolved or applied value may be checked before application or bound after application; its **realized** value is never a prerequisite for a pre-application check.

MAP product realization and application-attempt outcome are independent. **succeeded** means the requested im-

mutable MAP product was realized. **failed** means the attempt terminated at an exact failure stage and retains cause, affected scope, and nonmutation evidence without a realized MAP product or completion marker. **not_produced** means no MAP product was realized without asserting producer failure; the authorized empty-support case uses this value. A response or covariance status of **unavailable** inside a permitted realized base bundle is a companion-status fact, not attempt failure. Evidence-artifact realization is another independent axis.

For the r0.4 generation, the proposed owner-governed MAP profile semantics are **candidate**; scientific-owner approval is **pending**; canonical Registry registration is **pending**; source binding is **pending**; SCI-VAL activation/evaluability is **unavailable**; and an object-specific decision realization is **unavailable**. In particular, `SCI-MAP:observation_coadd_admission@2` is the proposed owner-governed aggregate-profile identity for this candidate. After exact owner acceptance, canonical Registry/source binding, and activation, SCI-VAL may evaluate it for a requested object. The same candidate-generation separation applies to `SCI-MAP:map_upstream_admission@2`. Earlier exact profile approvals and evaluations retain their own immutable source generations; this candidate does not rebind or relabel them.

Exact empty-support disposition. The 2026-09-08 owner directive selects for this r0.4 candidate the following empty-support disposition; final acceptance of these exact r0.4 bytes remains pending. If a valid applied support policy yields $\mathcal{S}_{\text{out}} = \emptyset$, or a valid applied coadd support policy yields $\mathcal{S}_{\text{out}}^c = \emptyset$, the application state is **applied_no_support_authorized_output_rows**, the product and attempt outcome is **not_produced**, and the exact cause is **no_support_authorized_output_rows**. The record retains request, effective, observation-resolved, and applied stages; exact parent and plan; coefficient, projection, support, role and profile identities; Q_* , thresholds and support predicates; proof that the output-row set is empty; attempt identity, cause, and provenance. It has no realized MAP product or completion marker and publishes no dense zero map, zero-sky measurement, valid ordinary signal plane, no-row coadd member, or fabricated response/covariance plane. No immutable empty-row scientific-bundle alternative is authorized.

2 Evidence record

Each requirement record contains at least:

Field	Required content
Requirement	Stable <code>SCI-MAP-REQ-NNN</code> and shared-authority digest.
Candidate	Exact source/build identity and environment profile.
Observable	Public product, typed state, API boundary, metadata, diagnostic, or failure transition under test.
Fixture	Immutable input identity and independent expected-value calculation.
Comparison	Exact predicate or preregistered numerical/statistical bound.
Result	Exactly one of pass , fail , blocked , not_applicable , or not_assessed . Skipped required evidence is not_assessed , never a pass.
Artifact realization	Exactly one of complete , incomplete , failed , or not_produced ; this records the evidence artifact and never substitutes for the requirement result.
MAP attempt outcome	Exactly one of succeeded , failed , or not_produced , independently of evidence-artifact realization.
MAP product realization	Exact immutable product identity and all five lifecycle stages only when actual accumulators, scientific rows, values, role-qualified statuses, and provenance were produced; otherwise absent.
Failure/nonproduction record	Reached lifecycle prefix, attempt identity, stage/cause/scope, provenance, and nonmutation evidence. A failure record does not realize a MAP product; no-support nonproduction is not failure.
Artifacts	Machine-readable outputs, logs, traces, plots/tables where needed, and digest-addressed provenance.
Reviewer	Independent reviewer identity, date, and deviation disposition.

Every ECS requirement result is exactly one of **pass**, **fail**, **blocked**, **not_applicable**, or **not_assessed**. **pass**

means complete positive evidence establishes the requirement for the exact candidate. **fail** means evaluated evidence contradicts it. **blocked** means evaluation cannot proceed because one exact named authority, dependency, profile, source binding, or prerequisite is unavailable. **not_applicable** means the requirement is outside the exact candidate’s declared role under an owner-approved rationale. **not_assessed** means applicable evidence has not yet been executed or reviewed. Evidence-artifact realization is a different axis with values **complete**, **incomplete**, **failed**, and **not_produced**.

Failures are retained as evidence. A tolerance, fixture, expected result, or applicability decision is not changed after viewing a failure without opening a new preregistered record.

3 Claim-layer gates

Passing one layer grants no stronger layer automatically.

Layer	Minimum engineering record	Prohibited inference
Algebraic contract	Exact owner-dispositioned shared authority and internally consistent independent expected values.	No claim that a candidate implements it.
Engineering conformance	Every applicable requirement passes on one exact candidate; no required skip, unresolved deviation, or false completion.	No claim that persisted WCS/response is scientifically faithful in real data.
Representation/response fidelity	Round-trip and injection evidence covers WCS, units, response classes, boundaries, and covariance representations with preregistered bounds.	No observational photometric, astrometric, or noise performance claim.
Observational performance	Independent references and repeatability establish the named amplitude, shape, position, covariance, or false-rate metric.	No production-readiness claim.
Production readiness	Operational profile, complete products, failure/log audit, dependency closure, release controls, and owner approval all pass.	No authority beyond the exact released profile and contract version.

All nine stable MAP scope decisions are decided for this candidate. A claim may still be blocked by an unapproved source binding, an unavailable producer, or missing conformance evidence; record that state without weakening the requirement or substituting a fallback.

4 Execution order and atomicity protocol

Run evidence in the following order so later evidence cannot mask an earlier contract failure:

1. **Stage A—structural.** Parse and freeze exact identity and parent, generation, source/profile binding, producer availability, and required decision realization for frozen PTC r0.5, the exact PTC-to-MAP boundary, and the AST same-*n* signal coordinate, all boundaries/profiles, units, lifecycle, WCS, MAP policy, and required-product plan. When the uniform route is selected, require the exact uniform Registry, family, and handoff identities declared in the shared authority; selection alone supplies no coefficient and creates no default. Reject direct CAL, PTC-disabled, missing-product, inferred no-op, and cross-generation routes before payload retrieval.
2. **Stage B—safe numerical classification.** Retrieve only structurally admitted transformed-signal and coefficient payloads. Classify z_i as finite, non-finite, or unrepresentable and γ_i as finite strictly positive, exact zero, negative finite, non-finite, or unrepresentable. Verify coefficient availability itself imposed no finiteness requirement. Separately validate the exact observation-resolved coverage cut before application: zero, ordinary (0,1],

above one admitted only by an independently authorized same-value/scope `coverage_cut_above_one_override_requested` fact, or invalid when missing, non-finite, negative, or unrepresentable. The number above one never authorizes itself. Invalid input fails before support construction, application, or mutation; no realized cut value is a pre-application prerequisite.

3. **Stage C—coordinate, product, and VAL decision.** Evaluate exact AST coordinate validity, one-hot placement/boundary, and only the companions required by the exact selected observation/coadd product role. To produce a new VAL decision, first verify exact profile semantics, scientific-owner approval, canonical Registry registration and source binding, activation/evaluability, and every required object fact. The current status is given by the exact candidate profile-status rule and permits no evaluation. After an exact active generation exists, perform the evaluation; then verify the resulting decision-artifact identity, request, applicability, eligibility, realization, exact causes, immutable reason, and failure scope. Only after that verification may MAP consume the decision for admission. When consuming an existing decision, verify its exact profile/source binding, object identity, four-axis state, causes, and realization before MAP admission. Only requested/applicable/eligible/realized passes. Verify separately that the PTC coefficient/QC pass uses its exact PTC-owned profile and neither replaces nor rescues `SCI-MAP:map_upstream_admission@2`.
4. **Stages D and E—membership then accumulation.** Construct membership only after Stages A–C and perform no payload arithmetic until all required gates pass. Verify no numeric-zero mask, late masking, or aggregate sanitization substitutes for admission. Then compare exact scientific output-row identity and cardinality, plus one-pixel, one-hot half-open-boundary, prohibited-fractional, empty-support, covariance, fixed-state response, full-procedure comparison-space typing, and distinct-fact observation outputs, to independent expected values; retained storage rows do not count as scientific output rows.
5. Exercise original-footprint exposure fixtures: one original influencing several descendants at different pixels; original and descendants in different pixels; missing original coordinate; boundary loss; multiple maps sharing one original; distinct observations with numerically identical original coordinates; and donor/synthesized descendants. Verify exact observation/UID/native/stable-ALIGN-slot/generation keys, layered AST ALIGN-grid parent in the target WCS, one own-coordinate placement, separate influence edges, and no precision or temporal-support promotion.
6. Verify complete immutable observation-bundle construction as a logical artifact. Independently request all three observation roles and record each reached requested, effective, observation-resolved, applied, and realized plan stage and separate attempt outcome; a successful realization has all five, a pre-application stop has no applied/realized records, and a failed or no-support not-produced applied attempt has no realized product. Exercise required publication for persisted routes plus explicitly selected in-memory and coadd-only routes. In-memory retention is not a waived product.
7. Snapshot coadd-owned state; test every compatibility failure under each selected role; require atomic nonmutation. Confirm that base-permitted unavailable or basis-incompatible response and incomplete covariance preserve the base signal membership and publish exact status/cause.
8. Confirm that the candidate observation-coadd profile cannot be evaluated before exact owner acceptance, Registry/source binding, and activation. After an active successor exists, evaluate all four object-decision axes separately from the MAP lifecycle and attempt outcome; then independently request all three coadd roles and exercise admitted centered-integer coaddition, exact coadd output-row identity, and companion placement. If a canonical target grid was requested before observation-map construction, verify that every constructed map honors it. Otherwise verify that already existing compatible bundles, including persisted coadd-only inputs, may select a common coadd target and use exact centered-integer embedding without modifying or relabeling their scientific samples. An explicit crop creates a new row-selected product with its own identity and provenance.
9. Round-trip all required representations and reconstruct each row-selected operator from provenance. Evaluate the physical FITS and typed WCS on every preregistered scientifically addressable pixel center and every outer-footprint cell vertex, including all corners, with the exact index mapping, common WCS generation/frame, binary64 great-circle metric, and fail-on-invalid rule. Infer no continuum maximum from that finite population.
10. Exercise fixed-state, PTC full-procedure, and PTC+MAP re-resolved response families. For every numerical Δz or Δm , require the complete baseline/perturbed identities, quantity/unit, rows, frames/WCS, reference/gauge/null state, supports, coefficient/response bases, exact common domain and both maps, unmatched-row treatment, perturbation parameters, finite-difference convention, state transition, availability, and cause. Reject implicit fill, intersection, coordinate/shape/position association, interpolation, reprojection, or gauge alignment; a state delta alone never authorizes subtraction. Require the stable `SCI-MAP-FULL-PROCEDURE-COMPARISON-SPACE` identity in every such record. Reconstruct and disclose the complete MAP-local operator $A_{\text{MAP},\Pi}$, the immutable realized plan Π , and row selector J_{out} with its exact output-row domain for every valid product, while marking coadd-full-procedure/re-resolved and whole-chain RTC-to-CAL-to-PTC-to-MAP responses unavailable unless separate

- approved authorities establish them; exercise an already realized PTC-grid companion applied exactly once; and honest response/covariance disclosure. For a response-bearing role, require only every exact response object named by that role; absence of a different unselected response family does not block it. For a response-bearing coadd, require one exact shared family R , compatible source/perturbation/basis/unit/normalization/gauge/WCS/row identities, fixed membership/placement/ B_{out} /support/coefficients, and an exact composition theorem; reject multiplication of a finite difference through B_{out} as a Jacobian without that theorem. For coadd covariance, require every block named by the selected covariance-qualified role before accepting the complete numerical propagation and otherwise require partial, symbolic, summarized, lineage-resolvable, or unavailable state on the base role. Verify no stronger role subsets an existing base parent and verify versioned binding for later derivative products.
11. Exercise consumer claim boundaries and immutable base/unfiltered MAP parentage without assuming a closed-world registry of all future scientific uses. Verify that Pointing and OOF receive only permission to reuse the exact ordinary MAP gridding operator when their inputs conform. Their WCS/frame, model/fitting, interpretation, outputs, defaults, profiles, and qualification remain mode-owned, and no unsupplied mode profile is activated.
 12. Repeat supported execution modes under the preregistered floating-point policy, while requiring exact discrete facts.
 13. Only after deterministic gates pass, execute representation, external, observational, and operational evidence appropriate to the requested claim.

For every failure-injection test, capture pre-state, attempted input, emitted diagnostic, post-state, publication/cardinality state, and completion state. No partial-state comparison may omit a coadd companion merely because its payload was not the injected failure target.

4.1 Scope-discrimination cases

The following cases make the targeted r0.4 amendments discriminating. Each case records every reached requested, effective, observation-resolved, applied, and realized state plus the separate attempt outcome, and asserts role-qualified atomic nonmutation on rejection.

The six independently requestable product roles are `SCI-MAP:base_observation_map@1`, `SCI-MAP:response_bearing_observation_map@1`, `SCI-MAP:covariance_qualified_observation_map@1`, `SCI-MAP:base_coadd@1`, `SCI-MAP:response_bearing_coadd@1`, and `SCI-MAP:covariance_qualified_coadd@1`. Every request has its own immutable identity, parent set, ordered MAP lifecycle, result, and failure scope. A stronger role is a new product over the exact base parent; it does not rewrite that parent's signal membership, response/covariance status, validity, or provenance.

Role	Exact parent	Required companions	Missing/incompatible companion
Base observation	Exact realized PTC/AST inputs and complete MAP-local observation plan	Complete MAP-local realized operator identity; honest upstream-response and covariance states	Base signal may remain valid; state and cause are retained.
Response-bearing observation	Exact base-observation product named above	Every named numerical response, with exact source domain, basis, class, unit, normalization, parent, WCS/grid, and row identity	This role fails or is unavailable before its required-product mutation; its base parent is unchanged.
Covariance-qualified observation	Exact base-observation product named above	Every within-observation covariance block named by the role on exact rows	Missing blocks are unknown and block this role only; its base parent is unchanged.
Base coadd	Ordered set of exact complete base-observation bundles and one resolved coadd plan	Every base signal companion and honest response and covariance identities/statuses	Unavailable or basis-incompatible response and incomplete covariance are permitted with unchanged signal membership; coadd response becomes unavailable with exact cause.

Role	Exact parent	Required companions	Missing/incompatible companion
Response-bearing coadd	Exact base-coadd parent named above and the identical ordered signal-member set	One exact response-family identity R ; every member response $R_o^{(R)}$; compatible source domain, perturbation definition, basis, class, unit, normalization, reference/gauge/null state, parent, WCS/grid, and row identity; fixed membership, placement, B_{out} , support and coefficients; and exact linear-composition authority	If the base parent already exists, this stronger role is unavailable for that parent; it never selects a subset. A separately requested admission may reject an observation before mutation only while constructing its own separately identified base/signal parent. No hidden subset or zero fill.
Covariance-qualified coadd	Exact base-coadd parent named above and the identical ordered signal-member set	Every named within- and cross-observation block on exact stacked rows	One missing required block blocks this role for that parent; no member subset is formed. A separately requested admission may reject before constructing its own identified base/signal parent. Unknown is never zero or independence.

Decision surface	Admitted case	Required discriminator
Uniform coefficient authority	Exact Registry/family/handoff selection and a PTC-complete same-generation payload.	Admit only through the frozen PTC-to-MAP boundary; no default, fallback, or coefficient synthesis.
Coverage cut	Finite requests at 0, within $(0, 1]$, and above 1.	Classify zero and ordinary values separately from the independently authorized typed above-one override; reject numerical self-authorization and missing, negative, non-finite, or unrepresentable values before support/application/mutation.
Coverage stages	Values differ across reached lifecycle stages.	Validate $c_{observation_resolved}$ before application, then require exact $c_{applied} = c_{observation_resolved}$ at the event. Evaluate the inclusive predicates with $c_{applied}$; create no future-stage placeholder and never require $c_{realized}$ to authorize application.
Coverage order statistic	$N = 0$ and preregistered small populations around integer transitions.	Reproduce the exact zero-based $\lfloor ([0.75N] + N)/2 \rfloor$ index and selected ordered value; for $N = 0$, retain exact $Q_* = 0$ without a selected population element or zero-sky meaning; verify inclusive thresholds and finite-positive row requirement.

Quantity-specific zero	Fixtures A–D: cancelling admitted signals, a constant-zero signal on nonempty support, index zero and placement offset zero, and separately zero normalization plus empty applied support.	Admit finite $N_p = 0$ when $Q_p > 0$ and every support/gate condition passes; accept zero index/offset only in their exact domains; keep $Q_p = 0$ unsupported, zero coefficients noncontributing, $c = 0$ unchanged, and empty support not_produced with its exact cause. No generic zero-invalid predicate is permitted. Record prospective state applied_no_support_authorized_output_rows , outcome not_produced , cause no_support_authorized_output_rows , complete support proof and applied provenance; produce no product, failure, completion marker, dense zero, or no-row coadd member.
Empty support	A valid authorized above-one cut produces no observation or coadd support-authorized rows.	The logical bundle and plan remain complete; publication is required only for routes whose plan requires persistence.
Bundle completeness	Required companions retained in memory, persisted, and consumed by a coadd-only route.	Honor the requested grid; otherwise select a compatible common coadd target and use exact centered-integer embedding. Reject relabeling or resampling as a way to manufacture compatibility.
Target grid	A preconstruction canonical grid is requested, or existing bundles have otherwise-compatible shapes.	Emit a new identified product with selector/provenance; do not mutate or relabel the base grid.
Crop	Explicit row selection after construction.	Disclose exact MAP-local A_{MAP}, Π separately from any upstream or whole-chain claim; numerical procedure deltas require both exact maps into one authorized
Response	Valid fixed-state and procedure-response products.	SCI-MAP-FULL-PROCEDURE-COMPARISON-SPACE domain and all gauge/support/perturbation facts.
Coadd response family	Same-family fixed responses, mixed families, and a finite difference without a composition theorem.	Bind exact family R and fixed coadd state for $R_{\text{coadd, out}}^{(R)} = B_{\text{out}} T_{\text{obs}}^{(R)}$; reject mixed families and Jacobian-like propagation without exact composition; keep coadd-procedure and whole-chain families unavailable. Record candidate , pending , pending , pending , unavailable , and unavailable across the six status axes and perform no object evaluation.
Profile activation	Candidate MAP/VAL profile bytes exist without canonical activation.	Preserve exact base parent and ordered membership; base-permitted unavailable companions do not reject; a selected required companion blocks only that stronger role and never forms a subset of an existing parent.
Product roles	Independently request base, response-bearing, and covariance-qualified observation and coadd roles.	

WCS population	Typed and FITS WCS differ at an interior center, outer boundary vertex, corner, and one invalid point in turn.	Evaluate the full preregistered population with exact spherical metric/frame/generation/ precision; reject on any invalid member and infer no continuum maximum.
NOI vocabulary	Present Q , formal precision, complete covariance, NOI conditional marginal second moment, NOI standardized signal, and an attempted significance label.	Preserve exact roles and claims; stronger physical-noise, uncertainty, significance, probability, false-rate, completeness, purity, or catalogue meaning remains unavailable without a separately approved method.
Pointing and OOF	Each requests exact ordinary MAP operator reuse.	Verify only the shared operator boundary; require separate mode-owned WCS/frame, model/fitting, interpretation, outputs, defaults, profile, and qualification, and leave an unsupplied mode profile unavailable.
Application status	Contract sources compile without an identified producer candidate.	Record contract completeness only; make no application availability or conformance claim.

SCI-NOI owns its exact realization-generation, empirical-uncertainty, conditional-scale, and standardization methods and products. Any physical- noise, total-uncertainty, covariance-completeness, significance, probability, false-rate, completeness, purity, or catalogue interpretation requires a separately approved exact method and remains unavailable by default. MAP therefore keeps Q normalization, formal conditional precision, complete conditional covariance, the NOI `conditional_detector_sign_randomization_marginal_second_moment`, the NOI standardized signal, and any future calibrated-significance method as distinct identities and claims.

5 Canonical requirement conformance clauses

The following clauses are generated from the shared authority. The first paragraph is the canonical science requirement; the second is its minimum conformance route. Every one of the 52 result fields is initialized to the exact value `not_assessed`; this says applicable evidence has not yet been executed or reviewed and does not describe artifact realization.

SCI-MAP-REQ-001 – Versioned authority

Canonical clause: The scientific contract version shall be v0.2. The rendered document revision shall be recorded separately as r0.4; changing layout or exposition alone shall not be represented as a new scientific contract version.

Minimum conformance evidence: Inspect both artifacts and their machine-readable provenance for distinct contract-version and document-revision fields.

Result: `not_assessed`

SCI-MAP-REQ-002 – Capability and realized PTC input

Canonical clause: The ordinary input shall be the realized SCI-PTC v0.1/r0.5 transformed calibrated- x product $z_i = Z_i^{\text{PTC}}$, a nonpolarimetric total-intensity-equivalent detector-time quantity in the inherited top-of-atmosphere, point-source-equivalent `mJy/beam` convention with its exact originating fixed-nominal-beam identity. Neither that unit nor a STOKES token establishes Stokes I. MAP has no CAL-to-MAP fallback, no inferred no-op PTC product, and no product when the required positive-rank PTC product is absent or PTC is disabled.

Minimum conformance evidence: Exercise exact realized PTC input, attempted CAL input, PTC-disabled and no-product routes, incompatible PTC generation, missing nominal-beam identity, and attempted Stokes-I promotion.

Result: `not_assessed`

SCI-MAP-REQ-003 – Exact occurrence identity

Canonical clause: Every candidate occurrence i shall bind observation, detector occurrence and UID, stable RTC output sample n , exact PTC product/application generation, segment, and array/network/group as applicable. Sample time is an associated attribute. Input-column, row, and container positions are locators only and shall not establish identity or equality.

Minimum conformance evidence: Permute times, containers, rows, and columns while preserving or breaking the exact parent chain; products shall follow exact occurrence identity and reject inferred joins.

Result: `not_assessed`

SCI-MAP-REQ-004 – Producer facts and MAP admission

Canonical clause: SCI-MAP shall consume PTC-owned transformed-signal availability, output-retention disposition, coefficient family/value/typed availability, and coefficient/QC proposition without reinterpretation. The PTC coefficient/QC proposition passes only when the coefficient is requested, applicable, eligible, and realized under the exact PTC-owned coefficient/QC profile for the declared MAP-facing family. It does not grant MAP use. SCI-MAP shall separately apply `SCI-MAP:map_upstream_admission@2` only after its exact owner acceptance, canonical Registry/source binding, and activation. [the exact candidate profile-status rule](#) applies to this generation. VAL Core may then evaluate that MAP-authored policy for an exact requested object but does not author it. The immutable r0.5 profile `SCI-MAP:map_upstream_admission@1` is historical, is not a compatibility alias, and shall not replace `SCI-MAP:map_upstream_admission@2`. Every fact, decision, cause, scope, and immutable reason remains separately typed.

Minimum conformance evidence: Audit the six profile/source/activation/decision axes; after activation, toggle each producer fact and each of the four named object-decision fields independently; verify exact PTC coefficient/QC projection, separate MAP admission, preserved causes, source/profile-version rejection, and no rescue or universal producer-flag action.

Result: `not_assessed`

SCI-MAP-REQ-005 – Coordinate and projection authority

Canonical clause: MAP shall consume `SCI-AST:rtc_output_grid_coordinates@1` for the same stable RTC sample n and complete parent chain as z_i ; it shall never join by time, row, cardinality, or coordinate equality. MAP owns G_{pi} through `SCI-MAP:one_hot_containing_pixel@1`: the unique finite in-grid pixel whose cell contains the coordinate under half-open bounds contributes $G_{pi} = 1$, every other pixel has $G_{pi} = 0$, and an outer-upper-boundary or out-of-grid coordinate contributes nowhere. Fractional projection is deferred. AST may materialize the relation only when the artifact binds the AST parent and exact MAP plan.

Minimum conformance evidence: Test exact-parent joins, centers, lower and upper cell boundaries, outer extent, invalid coordinates, sum-over-pixel normalization on its in-grid domain, and prohibited wrap/clamp/fractional placement.

Result: `not_assessed`

SCI-MAP-REQ-006 – Ordinary coefficient status

Canonical clause: The ordinary route requires one explicitly selected PTC-owned MAP-facing analysis/gridding coefficient family satisfying SCI-PTC-REQ-052-055 and SCI-PTC_TO_SCI-MAP_BOUNDARY v0.2-draft.1/r0.4. The historically owner-approved and source-bound Registry SCI-PTC:analysis_gridding_coefficients@draft-0.1 contains SCI-PTC:uniform_constant@draft-0.1, whose exact supported value is dimensionless $\gamma_i = \omega_i = 1$, and grants MAP permission through the historically owner-approved SCI-PTC:uniform_coefficient_handoff@draft-0.1. Those exact approvals and literal identifiers remain immutable; they neither activate this r0.4 candidate generation nor realize an object decision. Registration and permission do not select a family or realize a handoff. MAP shall use this family only after explicit selection; one immutable completed SCI-PTC v0.1/r0.5 transformed-product publication bound to the exact application, CAL parent, observation, and selection lineage and typed as a complete segment, complete scan, or bounded chunk wholly within one named scan or segment and one observation; exact target/payload/occurrence association; and a requested, applicable, eligible, realized profile evaluation under an exact active compatible binding generation. MAP does not reproduce, select, or infer the PTC-owned formula. An absent selection without an authorized mode default, unregistered family, inactive dependent binding, incomplete PTC publication, missing or mismatched payload, unavailable coefficient/QC decision, or incompatible generation makes the affected route unavailable; there is no unity or alternate-family fallback. Structural availability remains distinct from finiteness. MAP alone classifies an authorized payload as finite positive, exact zero, negative finite, non-finite, or unrepresentable and applies only a matching finite strictly-positive value. The uniform family supplies no precision, sensitivity, loading, scatter, response, covariance, exposure, validity, optimality, or default meaning.

Minimum conformance evidence: Audit retained uniform approval/source status separately from this candidate's activation/evaluability and object-decision state; then exercise registered-but-unselected, selected-but-unpublished, missing/mismatched association, nonpassing four-axis evaluation, incompatible generation, and exact active uniform cases; verify producer QC, MAP admission, numerical classification, contribution, and every identity, support, lifecycle, uncertainty, and prohibited meaning separately.

Result: not_assessed

SCI-MAP-REQ-007 – Exposure lineage and accounting

Canonical clause: ALIGN-owned e_{acq} and valid-original e_{vo} , in seconds, shall remain immutable on stable original occurrences through RTC/CAL/PTC. The products are normatively **upstream_eligible_original_footprint_exposure** and **retained_original_footprint_exposure**. The shorter prose labels “upstream-eligible exposure” and “retained exposure” are explicit aliases for those products and no others. MAP shall deduplicate by exact observation, detector occurrence/UID, stable original/native occurrence, stable ALIGN slot, and ALIGN mapping generation; place each positive e_{vo} original exactly once through SCI-AST_TO_SCI-MAP_ORIGINAL_FOOTPRINT_COORDINATE v0.1/r0.1, its layered AST ALIGN-grid direction/tangent/continuous-pixel parent in the exact target MAP WCS, and SCI-MAP:one_hot_containing_pixel@1; and never use a descendant RTC-output coordinate. Geometry/pointing realization, frame, coordinate validity, boundary state, parent, and version are required. Upstream-eligible and retained exposure differ only by ancestor-population restriction. Causal influence, representative RTC occurrence, and original-footprint exposure remain separate. These exposure products are not complete temporal support, effective map integration time, precision, or normalized-map influence. Missing original-coordinate authority makes the affected exposure role unavailable. Invalid-original has $e_{vo} = 0$; synthesis, replacement, donor reuse, overlapping filters, and decimation create no exposure. Boundary loss contributes nowhere. Coadd exposure unions atomically admitted observation-scoped original identities.

Minimum conformance evidence: Use one original influencing several descendants at different pixels, an original and descendants in different pixels, missing original coordinate, boundary loss, multiple maps sharing one original, distinct observations with identical coordinates, donor and synthesized descendants, and repeated lineage; verify exact own-coordinate placement, deduplication, and separate influence support.

Result: not_assessed

SCI-MAP-REQ-008 – Response information state

Canonical clause: Every valid SCI-MAP:base_observation_map@1 product shall disclose its known MAP-local realized operator response $A_{MAP,\Pi}$, including exact input/output domains, membership, projection, coefficient, support-row, WCS, parent, lifecycle, and limitations. MAP shall separately type upstream-conditioned fixed-state linear response, PTC full-procedure finite differences, and PTC+MAP re-resolved procedure response. An available fixed-state object shall also bind source domain, PTC-sample codomain, units, parents, response basis, and upstream limitations. Every numerical full-procedure Δz_{PTC-FP} , Δm_{PTC-FP} , or $\Delta m_{PTC+MAP-RR}$ shall bind the complete SCI-MAP-FULL-PROCEDURE-COMPARISON-SPACE record. A numerical full-procedure difference is defined only by the complete immutable SCI-MAP-FULL-PROCEDURE-COMPARISON-SPACE record. The record binds baseline and perturbed product identities; scientific quantity and unit; occurrence or MAP output-row identities; source and target frame/WCS; additive reference, gauge, and null- space state; baseline and perturbed support; coefficient and response basis; one exact common comparison domain; an exact map from each result into that domain; explicit treatment of rows present in only one result; perturbation amplitude, sign, side, location, source state, and normalization; finite- difference convention; state-transition record; and numerical availability with cause. It authorizes no implicit zero fill, union-domain sentinel, support intersection, coordinate-nearest or same-shape/row-position association, interpolation, reprojection, or gauge

alignment. Without the complete authorized record the numerical difference is unavailable, or the result is only a separately typed state transition/discontinuity. $\Delta\mathcal{S}_{\text{PTC-FP}}$ never establishes subtractability by itself. The PTC+MAP family reruns PTC and separately resolves Π_0 and Π_+ , whose membership and $\mathcal{S}_{\text{out}}$ may differ; subtraction is unavailable without exact maps into one authorized common comparison domain. A coadd-re-resolved/full-procedure response and a whole-chain RTC-to-CAL-to-PTC-to-MAP response each require separately authorized complete procedures and remain unavailable unless so established. Absent upstream response shall be reported as limited or unavailable without concealing the known MAP-local operator, invalidating an otherwise valid base map, fabricating a procedure/whole-chain identity, or prohibiting later analysis. A later response estimate or corrected map is a new immutable versioned product bound to the original MAP product and processing identity.

Minimum conformance evidence: Exercise MAP-local-known/upstream-unavailable, fixed-state, full-procedure common-domain and unavailable-domain cases, PTC+MAP re-resolved changing membership, companion-starting-at-MAP, partial, attempted coadd-procedure and whole-chain cases; verify exact typing, maps and rows, application exactly once, complete signal membership, honest limitations, and no fabricated identity or implicit subtraction.

Result: not_assessed

SCI-MAP-REQ-009 – Immutable plan, lifecycle, and attempt outcome

Canonical clause: the exact MAP lifecycle rule and the separate MAP attempt-outcome rule apply. The same ordered lifecycle shall bind the MAP plan, coefficient selection and application, projection, support policy, output plan, coadd plan, response and covariance requests, every independently requested product role, and coverage_cut. the exact lifecycle and independent-axis separation rule applies. For a valid applied policy with no support-authorized output rows, the exact empty-support disposition applies. Disabled expert values remain inactive; one observation's resolved state shall not leak into another.

Minimum conformance evidence: Compare serialized states across two deliberately different observations; stop one before application, fail one after application, and apply one valid above-one policy that yields no rows; verify ordered one-way provenance, separate attempt outcome, absence of fabricated realized products, exact failure evidence, exact prospective no-support disposition, axis separation, and complete replacement.

Result: not_assessed

SCI-MAP-REQ-010 – Factored contribution predicate

Canonical clause: For each pixel, membership in $\mathcal{C}_p$ shall be the conjunction of ten exact Boolean pass predicates derived from separately retained typed records: transformed-signal availability; PTC output retention; coefficient availability; PTC coefficient/QC proposition pass, γ_{QC} ; MAP upstream admission; AST coordinate validity; finite z_i ; finite strictly positive γ_i ; admitted projection/boundary relation; and MAP-local contribution plus required-companion predicates. Every VAL-evaluated gate retains named request, applicability, eligibility, and realization fields; only request=requested, applicability=applicable, eligibility=eligible, and realization=realized passes. Every other field combination is nonpassing with its exact cause, failure scope, and immutable reason retained. A structural source- or profile-binding failure is recorded only when that structural condition actually occurs; an ordinary nonpassing field value is not thereby a structural failure. Evaluation follows the ordered DAG: A structural identity/parent, generation, source/profile, producer availability, and decision-realization checks; B retrieval only of structurally admitted payloads and MAP classification of z_i and γ_i ; C AST validity, one-hot boundary/placement, and MAP-local required companions; D membership construction; E accumulation. All gates pass before payload arithmetic. Reading an authorized payload to classify it is not arithmetic, and excluded non-finite values never contaminate accumulation. T/F/U/C or four-field records shall never be conjoined directly; no numeric-zero mask represents exclusion.

Minimum conformance evidence: Vary one gate, each of the four named fields, source-binding failure, and knowledge state at a time; compare the typed record, Boolean projection, exact DAG stage, membership, cause/scope, and immutable reason; place NaN/Inf behind structurally excluded and admitted cases to distinguish safe classification from arithmetic contamination.

Result: not_assessed

SCI-MAP-REQ-011 – Ordinary accumulators

Canonical clause: On the fixed contribution set, SCI-MAP shall accumulate $N_p = \sum_i a_{pi} z_i$, $Q_p = \sum_i a_{pi}$, and, when available for the identical membership, $K_p = \sum_i a_{pi} k_i$, where $a_{pi} = G_{pi} \gamma_i > 0$.

Minimum conformance evidence: Use one-pixel and one-hot hand calculations and compare accumulators before normalization; fractional projection is not authorized in v0.2.

Result: not_assessed

SCI-MAP-REQ-012 – Normalized publication

Canonical clause: The scientific normalized output vector shall contain exactly the rows $p \in \mathcal{S}_{\text{out}}$ authorized by the applicable effective support policy. On those rows finite $Q_p > 0$ gives $\hat{m}_p = N_p/Q_p$, and an available response companion gives $\hat{k}_p = K_p/Q_p$. A finite exact-zero signal numerator $N_p = 0$ is admissible when all other gates and support facts pass

and gives the valid normalized scientific value $\hat{m}_p = 0$. No unsupported row, fabricated zero-filled substitute, or sentinel is a normalized scientific output or measured zero. If $\mathcal{S}_{\text{out}} = \emptyset$ after a valid applied support policy, [the exact empty-support disposition](#) applies.

Minimum conformance evidence: Exercise positive and finite exact-zero numerators on positive normalization, zero normalization, near-zero, overflowed, and non-finite aggregates plus a valid applied above-one cut yielding no rows; compare exact row identity and verify the no-support attempt produces no map or companion plane.

Result: not_assessed

SCI-MAP-REQ-013 – Accumulator interpretation

Canonical clause: The numerator is a pre-normalized accumulator and Q_p is a normalization fact by default. Neither shall be labeled a sky estimator, inverse variance, confidence, coverage, or significance without the separately required conditions.

Minimum conformance evidence: Audit schema, metadata, API types, and labels for every numerator and normalization representation.

Result: not_assessed

SCI-MAP-REQ-014 – Constant preservation

Canonical clause: With fixed admitted membership and positive coefficients, the ordinary estimator shall preserve a constant admitted sample input on every row in its exact support-authorized output set. This prediction shall not create rows outside that set or be promoted to compact-source, extended-source, integrated-flux, absolute-offset, or astrometric fidelity.

Minimum conformance evidence: Map constant fixtures with uniform and unequal coefficients; compare exact output-row membership, then confirm that stronger response claims remain separately gated.

Result: not_assessed

SCI-MAP-REQ-015 – Projection-specific estimand

Canonical clause: The authorized base projection is exactly `SCI-MAP:one_hot_containing_pixel@1`; it preserves unity over the finite in-grid domain and has explicit half-open boundary loss. Fractional projection, diagonal weighted least squares, and conservation across a finite boundary are outside this branch until separately authorized.

Minimum conformance evidence: Use containing-pixel and outer-boundary fixtures; attempt fractional or WLS substitution and require method rejection.

Result: not_assessed

SCI-MAP-REQ-016 – One exact MAP operator for linear companions

Canonical clause: For one selected immutable realized plan Π , $A_{\text{MAP},\Pi} \equiv A_{\text{out}} \equiv D_{Q,\text{out}}^{-1} J_{\text{out}} G \Omega$ is the only MAP operator and its complete realized identity is the MAP-local operator-response disclosure. Its identity binds the exact admitted PTC-sample input domain, support-authorized output-row domain, coefficient family/generation, one-hot projection plan, support policy, target WCS/grid, immutable parents, and lifecycle generation. Signal, fixed-state response, conditional covariance, an available realized PTC-grid response companion, and fixed noise realizations shall use that same operator, membership, placement, coefficient state, and output rows. A PTC full-procedure difference with MAP state fixed may use the same operator only after its baseline and perturbed products, common PTC-sample comparison domain, exact input maps, common output-row comparison domain, and exact output maps are authorized; equality to $A_{\text{MAP},\Pi} \Delta \mathbf{z}$ requires those exact common domains rather than array shape. A PTC-grid companion begins at MAP and receives the operator exactly once, never $H_{\text{fixed},\Theta}$ again. Physical original-footprint exposure instead uses unique originals and each original's own AST ALIGN-grid coordinate; causal influence is not a linear exposure companion.

Minimum conformance evidence: Trace all named uses coefficient by coefficient; compare operator, comparison-domain, row-map, and parent identities; verify complete MAP-local disclosure and exactly-once application; reject shape-, intersection-, interpolation-, and position-inferred differences; and separately trace original-footprint placement.

Result: not_assessed

SCI-MAP-REQ-017 – Realized response claim

Canonical clause: A stored kernel may be called a realized fixed-state response to template τ only when its exact sample parent satisfies $\mathbf{k} = H_{\text{fixed},\Theta} \tau$ and the support-restricted map satisfies $\hat{\mathbf{k}}_{\text{out}} = A_{\text{MAP},\Pi} H_{\text{fixed},\Theta} \tau = R_{\text{fixed},\Theta} \tau$ on exactly the signal rows. A PTC full-procedure finite difference $\Delta \mathbf{z}_{\text{PTC-FP}}$ and either map difference shall retain its distinct family identity and the complete comparison-space record, including both product identities, quantity/unit, rows, frames/WCS, reference/gauge/null state, supports, bases, common domain and exact two maps, unmatched-row treatment, perturbation and finite-difference convention, state transition, availability, and cause. $\Delta \mathbf{S}_{\text{PTC-FP}}$ records state change but never authorizes numerical subtraction. Otherwise the response is a tracer, separately typed discontinuity, limited, or unavailable.

Minimum conformance evidence: Inject a declared fixed-state template, comparable and noncomparable PTC full-procedure perturbations, a PTC+MAP re-resolved perturbation with different Π and rows, and a mismatched tracer; verify

family, complete comparison record, exact maps and rows, state differences, and no implicit fill, intersection, association, interpolation, reprojection, or gauge alignment.

Result: not_assessed

SCI-MAP-REQ-018 – Absolute-offset mode

Canonical clause: SCI-MAP v0.2 shall apply no signal-centering operation: $L = I$. Coaddition shall not subtract a mean, remove a constant mode, or recenter a source; any upstream loss of that mode shall remain explicit in response and provenance.

Minimum conformance evidence: Use a constant-offset fixture and inspect both the mapped value and recorded upstream response/null-mode state.

Result: not_assessed

SCI-MAP-REQ-019 – Conditional covariance

Canonical clause: When a declared PTC-domain covariance N_{PTC} is available for the exact fixed membership and state Θ , MAP shall propagate it as $C_{m,\text{out}} = A_{\text{MAP},\Pi} N_{\text{PTC}} A_{\text{MAP},\Pi}^T$, retaining consequential off-diagonal terms on the exact operator domains. The equation is complete only on the declared available domain. Missing, limited, symbolic, summarized, lineage-resolvable, or unavailable terms remain typed and are never zero-filled or interpreted as independence; absence of a complete model does not invalidate the signal map.

Minimum conformance evidence: Compare a small exact propagation and honest partial, symbolic, summarized, lineage-resolvable, and unavailable states on identical operator and row membership.

Result: not_assessed

SCI-MAP-REQ-020 – When normalization is precision

Canonical clause: Normalization may be called marginal inverse variance only when the exact selected γ family is calibrated true inverse variance, selected errors are independent on the asserted domain, and $\sum_i \gamma_i (G_{pi}^2 - G_{pi}) = 0$. One-hot geometry alone does not prove the coefficient premise.

Minimum conformance evidence: Test one-hot geometry with both proven inverse-variance and nonprecision coefficient families.

Result: not_assessed

SCI-MAP-REQ-021 – Formal diagonal weight

Canonical clause: When its assumptions hold, $\phi_p = Q_p^2 / \sum_i G_{pi}^2 \gamma_i$ is the formal marginal inverse variance. It remains distinct from Q_p , support-restricted precision $C_{m,\text{out}}^+$, exposure, hits, and empirical statistical weight.

Minimum conformance evidence: Compute and label every object for an authorized one-hot fixture and a correlated counterexample.

Result: not_assessed

SCI-MAP-REQ-022 – Covariance and nuisance disclosure

Canonical clause: The bundle shall report what uncertainty or covariance information it actually provides, its meaning, domain, assumptions, and limitations, and whether fuller information is persisted, lineage-resolvable, summarized, incomplete, or unavailable. It shall list omitted correlations and nuisance terms; absence of a complete covariance model shall not invalidate the map or prohibit later analysis, and unavailable information shall not be set to zero. A later covariance estimate may be attached only as a new versioned product bound to the original MAP product and processing identity.

Minimum conformance evidence: Inspect every representation state, prevent promotion of normalization, hits, or diagnostic weight to covariance, and verify derivative-product binding without rewriting the original claim.

Result: not_assessed

SCI-MAP-REQ-023 – Data-derived state

Canonical clause: If membership, coefficients, response, or geometry are estimated or selected from the same data, the displayed linear expectation and covariance shall remain explicitly conditional; unconditional bias or covariance requires a named joint model or resampling result outside this fixed-state claim.

Minimum conformance evidence: Use metadata cases for fixed and data-derived state and verify that unconditional labels appear only with separately admitted evidence.

Result: not_assessed

SCI-MAP-REQ-024 – No significance promotion

Canonical clause: Signal multiplied by the square root of merely formal weight shall be labeled formal standardized signal, not sig2noise or empirical significance. SCI-NOI owns its exact realization-generation, empirical-uncertainty, conditional-

scale, and standardization methods and products. Any physical- noise, total-uncertainty, covariance-completeness, significance, probability, false-rate, completeness, purity, or catalogue interpretation requires a separately approved exact method and remains unavailable by default. MAP therefore keeps Q normalization, formal conditional precision, complete conditional covariance, the NOI `conditional_detector_sign_randomization_marginal_second_moment`, the NOI standardized signal, and any future calibrated-significance method as distinct identities and claims. In particular, MAP shall not equate this quantity with NOI's conditional marginal second moment or standardized signal, and no future calibrated significance product is inferred.

Minimum conformance evidence: Attempt each prohibited relabeling through all public representations; compare Q , formal conditional precision, complete conditional covariance, the exact NOI marginal-second-moment and standardized-signal roles, and a hypothetical separately authorized significance role; require distinct identities and claims.

Result: `not_assessed`

SCI-MAP-REQ-025 – Distinct map facts

Canonical clause: Every base/unfiltered MAP bundle shall separately represent AST-geometric incidence, MAP-route-candidate incidence, estimator-admitted contribution, admitted-observation count, `upstream_eligible_original_footprint_exposure`, `retained_original_footprint_exposure`, numerical-normalization support, science-policy support, and MAP-local base-product scientific validity. No fact or downstream eligibility decision shall become authority for another.

Minimum conformance evidence: Construct cases in which each pair differs and verify separately typed identities and immutable ownership.

Result: `not_assessed`

SCI-MAP-REQ-026 – Incidence layers

Canonical clause: AST-geometric incidence records valid coordinate placement independent of signal availability; MAP-route-candidate incidence additionally requires the exact realized PTC route and occurrence-level upstream admission; estimator contribution additionally requires all remaining gates in C_p . The one-hot touch convention counts the unique containing pixel only.

Minimum conformance evidence: Use coordinate-valid/signal-unavailable, route-ineligible, zero-coefficient, and admitted occurrences and compare all three populations.

Result: `not_assessed`

SCI-MAP-REQ-027 – Estimator contributions

Canonical clause: Estimator-admitted contributions shall count exactly $i \in C_p$. A candidate ineligible under the exact MAP admission or contribution policy for its preserved cause, or with unavailable/conflicting required state, zero/negative/non-finite coefficient, non-finite signal, or out-of-grid projection, contributes nowhere; no producer flag implies a universal action.

Minimum conformance evidence: Compare exact counts with an independently evaluated profile and factored-gate oracle while preserving cause and decision scope.

Result: `not_assessed`

SCI-MAP-REQ-028 – Admitted observation count

Canonical clause: For coadds, admitted observation count shall count complete observation bundles contributing at the pixel and shall not be confused with sample hits or observation attempts.

Minimum conformance evidence: Use one complete, one unsupported, and one rejected observation bundle and verify the per-pixel count.

Result: `not_assessed`

SCI-MAP-REQ-029 – Upstream-eligible original-footprint exposure

Canonical clause: MAP `upstream_eligible_original_footprint_exposure` shall union and deduplicate exact positive- e_{vo} original occurrences that are ancestors of route candidates admitted by SCI-MAP:`map_upstream_admission@2`. Each original is placed once by its own exact AST coordinate and the one-hot half-open rule; it is not placed at every descendant coordinate. The exposure population precedes coefficient positivity and MAP-local numerical rejection. Missing original-coordinate or lineage authority makes this exposure role unavailable.

Minimum conformance evidence: Use one original influencing several candidates at different pixels, plus an eligible zero-coefficient candidate; verify one own-coordinate exposure placement, separate influence edges, and no precision promotion.

Result: `not_assessed`

SCI-MAP-REQ-030 – Retained original-footprint exposure

Canonical clause: MAP `retained_original_footprint_exposure` shall use the same original-occurrence key and own-coordinate placement as REQ-029, restricted to originals that are ancestors of estimator-admitted members. Overlapping filters, decimation, replacement, synthesis, and multiple descendants do not duplicate seconds. Coadded exposure shall union observation-scoped original identities after atomic coadd admission; one original can occupy at most one pixel under the base one-hot family, or none after boundary loss. Neither exposure is precision.

Minimum conformance evidence: Use overlapping observation lineage, multiple descendants, mixed admission, donor reuse, and boundary loss; compare exact original-key unions, pixel placement, and coadd sums.

Result: `not_assessed`

SCI-MAP-REQ-031 – Adopted threshold selector

Canonical clause: For each declared support-policy population, sort its finite strictly positive normalization coefficients Q_p ; for count $N > 0$, select zero-based $k = \lfloor ([0.75N] + N)/2 \rfloor$ and $Q_\star = Q_{(k)}$; for $N = 0$, use the exact boundary convention $Q_\star = 0$, which selects no population element and never denotes a zero-sky measurement. The exact population and policy identities and each reached requested, effective, observation-resolved, applied, and realized state/value of dimensionless $c = \text{coverage_cut}$ shall be retained; a successful realized product contains all five, while failure or nonproduction retains only the reached prefix. Pre-application admission shall check the exact observation-resolved value. Application shall bind that value to one support-policy attempt; realization shall later bind only an actual product. The exact Boolean plan fact `coverage_cut_above_one_override_requested` shall bind request identity, authorized owner or profile, scope, each reached lifecycle stage, exact value, cause, and provenance. It has no applied/realized record before application and no realized record after failure or nonproduction. The scalar shall be finite and nonnegative. Exact zero requests no relative cut while finite strictly-positive Q_p remains mandatory. Values $0 < c \leq 1$ are the ordinary recommended range. A value $c > 1$ is admissible only when that independently authorized fact is true for the same resolved value and scope; the number cannot authorize itself and may yield no support-authorized output rows. That prospective case is governed by [the exact empty-support disposition](#). Negative, non-finite, missing, or unrepresentable values shall fail before support-row construction, application, or required-product mutation. No universal numerical default is established.

Minimum conformance evidence: Check $N = 0$ with exact $Q_\star = 0$ and no selected element, and small N values around every integer-index transition against the exact zero-based formula; verify each reached stage, all five only on successful product realization, pre-application use of the resolved value, absence of future-stage placeholders, separate failed and not-produced outcomes, exact-zero and ordinary endpoints, independently authorized above-one override with same-value/scope matching, self-authorization rejection, prospective no-support nonproduction, and preconstruction nonmutation for every invalid state.

Result: `not_assessed`

SCI-MAP-REQ-032 – Separate support predicates

Canonical clause: Only after the effective plan and exact observation-resolved cut pass REQ-031 shall application bind $C_{\text{applied}} = C_{\text{observation_resolved}}$ for one support-policy attempt. Numerical normalization support shall then require finite positive $Q_p \geq Q_\star C_{\text{applied}}/10$, and science-policy support shall independently require finite positive $Q_p \geq Q_\star C_{\text{applied}}$. Both comparisons are inclusive and use the exact declared population and inherited order-statistic rule, including $Q_\star = 0$ when $N = 0$. At $c = 0$, no relative threshold excludes a row, but no zero, negative, non-finite, or absent Q_p is restored. An authorized expert $c > 1$ may leave the science-support set empty; [the exact empty-support disposition](#) governs that applied attempt. These thresholds are an operational relative-support policy and imply no exposure, precision, significance, response, completeness, or physical optimality. Neither support state establishes final validity or precision; no realized value is required to authorize this pre-application check.

Minimum conformance evidence: Exercise values below, exactly at, and above both thresholds for empty and nonempty populations; distinguish the $N = 0$, $Q_\star = 0$ boundary convention from any selected element or zero-sky measurement; distinguish normalization from science support; verify resolved-to-applied equality, exact-zero behavior, no self-authorizing above-one value, prospective no-support nonproduction, and no use of C_{realized} as a prerequisite.

Result: `not_assessed`

SCI-MAP-REQ-033 – MAP-local base-product validity

Canonical clause: One MAP-local base-product scientific-validity state shall require membership in the exact support-authorized row set, exact admitted PTC and AST parents, passing MAP contribution/support policy, finite normalized signal, and absence of an applicable MAP-product failure. Explicitly limited or unavailable response/covariance does not alone invalidate the map. An empty support-authorized row set realizes no MAP product and therefore has no base-product validity state; [the exact empty-support disposition](#) governs it. Retained storage, finite payload, coefficient, exposure, incidence, normalization, or downstream-local eligibility cannot establish or rewrite MAP validity.

Minimum conformance evidence: Use truth-table fixtures separating producer state, MAP admission, response/covariance disclosure, base validity, later use-local eligibility, and the prospective no-support not-produced outcome with no validity state.

Result: `not_assessed`

SCI-MAP-REQ-034 – Classify before arithmetic

Canonical clause: Only structurally admitted payloads shall be retrieved and safely classified for finiteness/sign before membership construction. All required gates shall pass before payload arithmetic. Multiplication by zero, masking after accumulation, or sanitizing a non-finite aggregate shall not substitute for admission.

Minimum conformance evidence: Place NaN and Inf payloads behind structural false and true predicates; verify no unauthorized read, exact numerical classification when authorized, no contamination, diagnostics, and failure scope.

Result: not_assessed

SCI-MAP-REQ-035 – Exceptional contribution states

Canonical clause: Zero coefficient shall be noncontributing, a low positive coefficient shall follow the explicit adopted policy, negative or non-finite ordinary coefficients shall be invalid, and an out-of-bounds projection shall contribute nowhere. Each cause shall remain distinguishable.

Minimum conformance evidence: Run one-cause-at-a-time fixtures and inspect cause-specific states plus unchanged neighboring pixels.

Result: not_assessed

SCI-MAP-REQ-036 – Complete observation bundle and roles

Canonical clause: The six independently requestable product roles are SCI-MAP:base_observation_map@01, SCI-MAP:response_bearing_observation_map@01, SCI-MAP:covariance_qualified_observation_map@01, SCI-MAP:base_coadd@01, SCI-MAP:response_bearing_coadd@01, and SCI-MAP:covariance_qualified_coadd@01. Every request has its own immutable identity, parent set, ordered MAP lifecycle, result, and failure scope. A stronger role is a new product over the exact base parent; it does not rewrite that parent's signal membership, response/covariance status, validity, or provenance. A SCI-MAP:base_observation_map@01 shall exist as one immutable ordered logical bundle before coaddition only when actual support-authorized scientific rows and product values were realized. It contains signal, numerator, normalization identity, complete MAP-local realized operator identity, honest upstream-response and uncertainty/covariance identity and status, the distinct facts in REQ-025, exposure lineage/accounting identity, unit and exact nominal-beam parent, complete AST frame/WCS, shape/indexing, estimator/projection/profile identities, PTC removed-component/additive-reference/null-space state, all five MAP lifecycle stages, successful attempt outcome, exact PTC/AST parentage, and product identity. It may remain valid when a stronger numerical upstream response or complete covariance is unavailable; those permitted companion statuses are not attempt failure. Independently requested SCI-MAP:response_bearing_observation_map@01 and SCI-MAP:covariance_qualified_observation_map@01 products shall each parent the exact realized base observation, bind their own lifecycle and attempt outcome, and require respectively every named exact response-family object or every named within-observation covariance block. An applied no-support attempt creates no logical observation bundle. Physical persistence is separately controlled by the effective output plan.

Minimum conformance evidence: Request each observation role independently; remove or alter each required identity/companion; distinguish absent declarations, honest unavailable companion states, failed attempts, and no-support non-production; verify role-specific failure without base-parent mutation; and verify identical scientific content and identity for in-memory and faithfully persisted forms.

Result: not_assessed

SCI-MAP-REQ-037 – Role-qualified atomic coadd admission

Canonical clause: After exact owner acceptance, canonical Registry/source binding, and activation make SCI-MAP:observation_coadd_admission@02 evaluable, a complete realized observation bundle shall pass every compatibility check and only the companion checks required by the exact selected product role before any coadd-owned state changes. This r0.4 candidate has no active profile or object-specific decision. Under SCI-MAP:base_coadd@01, unavailable or basis-incompatible response and incomplete covariance do not alter signal membership and do not reject an otherwise compatible base bundle; their exact identities, states, and causes remain. A required companion incompatibility is atomically decisive only for SCI-MAP:response_bearing_coadd@01 or SCI-MAP:covariance_qualified_coadd@01 as applicable. If an exact base coadd parent already exists, such an incompatibility blocks the stronger role for that parent and never selects a subset. A separately requested stronger-role admission may reject an observation only before constructing its own explicitly identified base/signal parent. Every rejection leaves all affected numerators, normalizations, counts, exposure, response, validity, provenance, and product cardinality unchanged.

Minimum conformance evidence: First verify all six profile-status axes and require no evaluation while activation is unavailable; after an exact active generation exists, snapshot all role-owned and base-parent state, inject one base fact, response-family, and covariance incompatibility at a time, and verify role-specific admission, exact pre-mutation rejection, unchanged base membership, and no companion-only subset.

Result: not_assessed

SCI-MAP-REQ-038 – Centered integer common grid

Canonical clause: A canonical target grid may be requested and made immutable before observation-map construction. Observation maps constructed on that exact grid, or otherwise compatible observation/coadd shapes, shall use exact centered-integer embedding: shape differences are nonnegative and even in each axis, and the corresponding integer offset maps reference pixels to the same world coordinate. A later crop is a newly identified row-selected product with immutable parent and lineage; it shall not relabel the parent or make it retrospectively compatible. Fractional shift, post-hoc pad, reprojection, interpolation, wrapping, implicit recentering, mosaicking, and a different sky grid are outside this estimator.

Minimum conformance evidence: Test preconstruction target-grid requests, every parity/sign case and WCS reference-pixel relation, exact integer embedding, a separately identified post-hoc crop, attempted parent relabeling, and each excluded grid-changing operation.

Result: not_assessed

SCI-MAP-REQ-039 – Ordinary coadd estimator

Canonical clause: The exact v0.2 family `SCI-MAP:uniform_observation_coadd_coefficient@1`, owned by MAP, assigns dimensionless $u_{op} = 1$ to every row of each bundle eligible and realized under an exact owner-accepted, canonically registered/source-bound, active `SCI-MAP:observation_coadd_admission@2` generation and realized object decision. [the exact candidate profile-status rule](#) applies to this generation. The VAL request/applicability/eligibility/decision-artifact-realization fields remain distinct from the five-stage MAP coadd lifecycle and attempt outcome. The coefficient is an equal-observation coefficient, not precision or inverse variance. Each of `SCI-MAP:base_coadd@1`, `SCI-MAP:response_bearing_coadd@1`, and `SCI-MAP:covariance_qualified_coadd@1` is independently requestable, has its own exact base parent, lifecycle, and attempt outcome, and uses identical centered-integer equal-observation signal arithmetic for its admitted ordered membership. On exact row identities, $N_p^c = \sum_o u_{op} \hat{m}_{op}$, $Q_p^c = \sum_o u_{op}$, and $\hat{s}_p = N_p^c / Q_p^c$ is published only on $\mathcal{S}_{out}^c$. If $\mathcal{S}_{out}^c = \emptyset$, no coadd is realized and [the exact empty-support disposition](#) applies. Conforming in-memory and persisted bundles have identical admission, identities, arithmetic, and result; storage route is not a scientific weighting or eligibility fact.

Minimum conformance evidence: Audit all six candidate profile-status axes; after activation, evaluate all four VAL fields separately from every MAP lifecycle/attempt field; independently request all coadd roles; use empty-support, one-observation, and two-observation cases from in-memory and persisted conforming routes; verify exact parent/membership, arithmetic equality, no product for empty support, and rejection of precision labels.

Result: not_assessed

SCI-MAP-REQ-040 – Family-qualified coadd response

Canonical clause: The stacked signal vector $\hat{\mathbf{m}}_{obs}$ shall carry explicit observation-row identities and B_{out} is the exact centered-integer equal-observation averaging operator. For one exact response family R , $T_{obs}^{(R)}$ is the ordered stack of the exact compatible member responses $R_o^{(R)}$, and $R_{coadd,out}^{(R)} = B_{out} T_{obs}^{(R)}$ only when every member has that same family identity; source domain, perturbation definition, basis, unit, normalization, reference/gauge/null state, WCS/grid and row maps are compatible; coadd membership, centered-integer placement, B_{out} , support and coefficient state are fixed for the query; and the family is mathematically composable through that fixed linear coadd operator. `SCI-MAP:response_bearing_coadd@1` binds the exact response-family identity it carries. Fixed-state, PTC full-procedure finite-difference, PTC+MAP re-resolved, coadd-re-resolved/full-procedure, and whole-chain responses are never one unqualified family. A finite difference is not multiplied through B_{out} as though it were a linear Jacobian without an exact composition theorem. A perturbation that may alter observation admission, coadd membership, support, placement, coefficients, role, or plan requires a separate coadd-full-procedure or coadd-re-resolved family and its own `SCI-MAP-FULL-PROCEDURE-COMPARISON-SPACE` record. Both that coadd-procedure response and the whole-chain response are unavailable until separately authorized. `SCI-MAP:base_coadd@1` requires every admitted member's complete base bundle and all response identities/statuses, but permits unavailable or basis-incompatible response with unchanged numerical signal membership. It shall then disclose coadd response unavailable with exact cause, never a hidden response subset, zero companion, or assumed compatibility. `SCI-MAP:response_bearing_coadd@1` parents one exact base coadd with identical ordered membership and requires the exact family identity and every member $R_o^{(R)}$ plus all fixed-state and composition conditions above. If the requirement fails for an existing parent, the stronger role is unavailable and the parent is unchanged. Any separately selected response-bearing admission that rejects a member shall construct a separately identified base/signal parent and reject before any of its state mutates. Exposure and count follow that selected admission, with exposure's unique-original placement.

Minimum conformance evidence: Use exact same-family fixed-state members, mixed families, changed perturbation definitions, incompatible bases/gauges/rows, a finite difference without a composition theorem, and separately unavailable coadd-procedure/whole-chain families; verify unchanged base membership, unavailable response with exact cause, exact stronger-role family/parent identity, no subset, and atomic admission for a separately identified request.

Result: not_assessed

SCI-MAP-REQ-041 – Coadd covariance and precision

Canonical clause: Σ^{obs} shall be the block covariance matrix on exact stacked observation-row identities with every available

within- and cross-observation block explicitly located. `SCI-MAP:base_coadd@1` requires covariance identities/statuses and permits partial, symbolic, summarized, lineage-resolvable, or unavailable state. `SCI-MAP:covariance_qualified_coadd@1` parents one exact base coadd with identical ordered membership and requires every named within- and cross-observation block before MAP may publish $C_{c,out} = B_{out} \Sigma^{obs} B_{out}^T$. Missing blocks are unknown and block only this stronger role; they are never zero or independence. If a block is absent for an existing parent, no subset is selected and the base coadd remains unchanged. Any separately selected covariance-qualified admission that rejects a member shall construct its own separately identified base/signal parent before mutation. The uniform observation coefficient is not precision.

Minimum conformance evidence: Use complete within/cross blocks and every missing-block state under both roles; verify exact propagation only for the complete stronger role, unchanged base membership and validity, no subset, and exact parent/role identity.

Result: `not_assessed`

SCI-MAP-REQ-042 – Correlated-GLS boundary

Canonical clause: A correlated minimum-variance solution using $(\Sigma_p^{obs})^{-1} \mathbf{1}$ may require negative combination coefficients and therefore belongs to a separately authorized correlated-GLS branch, not the ordinary positive-coefficient coadd. The token K_p remains exclusively the MAP response/kernel accumulator and never names a covariance matrix.

Minimum conformance evidence: Use positive- and negative-correlation covariance matrices that distinguish ordinary weighting from the GLS benchmark; require method and notation identity to remain explicit.

Result: `not_assessed`

SCI-MAP-REQ-043 – Minimum product and attempt identity

Canonical clause: Every realized observation and coadd product shall identify one exact product-role token from the six-role Registry; its independent request and all five MAP lifecycle stages; successful attempt outcome; exact base parent and ordered membership; external observation/coadd, array/group, occurrence, and row domains; estimator and coefficient families; MAP-local operator and upstream-conditioned response/covariance states; exact response-family identity for every response-bearing product; signal unit and fixed-nominal-beam parent; complete AST frame/WCS and WCS-generation identity; preconstruction target-grid request where applicable; grid/projection/boundary; exact active MAP profiles and object decisions; exact PTC product/application and coefficient generations; exposure convention; logical-bundle and plan-controlled persistence state; parentage; causes; and product identity. Any numerical full-procedure response additionally binds the complete `SCI-MAP-FULL-PROCEDURE-COMPARISON-SPACE` record; coverage policy binds its typed above-one override record. A failed or not-produced attempt instead binds only the reached lifecycle prefix, attempt identity/outcome, exact cause, affected scope, provenance, and nonmutation evidence and has no product identity or completion marker. No STOKES token supplies quantity authority. Scientific identity shall be durable without prescribing a persistence format.

Minimum conformance evidence: Round-trip every role, lifecycle, attempt outcome, parent, membership, response family, comparison-space, and override identity through prospective in-memory and persisted representations; detect collisions or generation mixing; distinguish unavailable companion status from attempt failure; and reject a bare token, storage route, or serialization convention as scientific authority.

Result: `not_assessed`

SCI-MAP-REQ-044 – Frames and units

Canonical clause: MAP shall bind the complete frame identity delivered by `SCI-AST:rtc_output_grid_coordinates@1`: exact source/target frame, reference system (including ICRS or FK5/J2000 as actually declared), equinox/epoch, time scale, site/model state, TAN topology, WCS, axis order, signs, handedness, and degree-valued science axes. MAP shall not equate J2000, FK5, ICRS, apparent, observed, or unrefracted frames by name or default. Signal/response units are `mJy/beam` with exact nominal-beam identity; covariance has its square.

Minimum conformance evidence: Inspect complete frame and unit/beam identity and require cross-frame, ambiguous-frame, cross-unit, or cross-beam coadd rejection.

Result: `not_assessed`

SCI-MAP-REQ-045 – WCS and index persistence

Canonical clause: Full-precision typed or sidecar WCS shall be the lossless admission and provenance authority. For an immutable finite MAP grid of n_x by n_y cells, the 0.1-arcsec representation-test population is the finite union of (a) every scientifically addressable integer pixel center $(j+1, k+1)$ for $0 \leq j < n_x$, $0 \leq k < n_y$, and (b) every outer-footprint cell vertex: $(h+\frac{1}{2}, \frac{1}{2})$, $(h+\frac{1}{2}, n_y+\frac{1}{2})$ for $0 \leq h \leq n_x$, together with $(\frac{1}{2}, l+\frac{1}{2})$ and $(n_x+\frac{1}{2}, l+\frac{1}{2})$ for $0 \leq l \leq n_y$. The second set includes all four outer corners and every required outer-boundary vertex. The complete population is preregistered from the full-precision typed WCS, grid shape, and row domain before the physical FITS WCS is generated. Both representations shall bind the same WCS-generation identity, celestial frame, axis order, signs, handedness, orientation, shape, and centered-integer reference-pixel relation. At every population member, each WCS is evaluated at the exactly corresponding center (zero-based memory index (j, k) , one-based FITS pixel $(j+1, k+1)$). Separation is the great-circle angle $\text{atan2}(\|\mathbf{u} \times \mathbf{v}\|, \mathbf{u} \cdot \mathbf{v})$ between unit direction

vectors in that exact declared frame, evaluated with IEEE-754 binary64 round-to-nearest, ties-to-even operations and reported in arcseconds. The maximum over the complete finite population shall be at most 0.1 arcsec. If either WCS yields an invalid or non-finite direction at any preregistered member, the representation fails; the point is not dropped. A center is absent only when its scientific row was already absent before population generation; every listed outer-boundary vertex remains required. No smaller sampled subset or continuum/global-sky claim is authorized. The 0.1-arcsec bound is only the maximum over that exact preregistered finite population; it is not a continuum or global-sky maximum. Exact axis sign, handedness, orientation, and centered-integer shape/reference-pixel relations remain required independently of the numerical bound.

Minimum conformance evidence: Evaluate every preregistered center and outer-boundary vertex with the stated index relation, frame, WCS generation, binary64 spherical metric, and invalid-point rule; calculate the exact finite-population maximum and separately verify every orientation invariant.

Result: `not_assessed`

SCI-MAP-REQ-046 – Lifecycle-bound provenance

Canonical clause: Every MAP plan, coefficient selection/application, projection, support policy, output plan, coadd plan, response/covariance request, product role, and coverage policy shall record each reached requested, effective, observation-resolved, applied, and realized identity in order and shall separately record application-attempt outcome. The record shall include each reached exact `coverage_cut` stage, with all five included only for successful product realization; the typed `coverage_cut_above_one_override_requested` identity, authority/profile, scope, value, cause, provenance, and binding to every reached stage; exact PTC/AST/boundary/profile and admitted coefficient Registry/family/handoff generations; the six profile semantics/approval/registration/source/activation/object-decision axes; every typed gate's independent request/applicability/eligibility/decision-artifact-realization fields plus Boolean projection, immutable reason, cause, failure scope, and DAG stage; exact product-role and base-parent identities; admitted contributions/bundles; the complete MAP-local operator $A_{\text{MAP},\Pi}$, accumulators, and row domains when realized; unique-original exposure facts; PTC additive-reference/removed/null states; exact response family, composition authority, and `SCI-MAP-FULL-PROCEDURE-COMPARISON-SPACE` or state-transition record as applicable; covariance state; WCS comparison population/generation; centered-integer offsets; persistence state; required products; failures; and not-produced causes. No applied or realized record shall be fabricated before application. A terminal failure retains the applied attempt and failure evidence without a realized product; an applied no-support outcome retains its exact proof and cause with `not_produced` outcome and no failure or realized product.

Minimum conformance evidence: Reconstruct each reached lifecycle stage, attempt outcome, role/parent, row-selected operator, cut/override transition, coefficient handoff, profile-status axis, exposure population, gate projection, response family/comparison-space map, WCS population, logical/persisted relation, and typed unavailable, failed, or no-support not-produced state from provenance alone.

Result: `not_assessed`

SCI-MAP-REQ-047 – Failure scope, nonproduction, and atomicity

Canonical clause: An invalid required input, unauthorized above-one cut, incompatible bundle, unrepresentable finite aggregate, unrepresentable coordinate/index conversion, missing companion required by the exact selected role, or failed required publication shall propagate at its declared contribution, pixel, observation-admission, or complete-product scope and shall fail before the affected live state changes. A pre-application failure creates no applied or realized record. A failure after application retains the exact applied attempt, failure stage, cause, affected scope, and nonmutation evidence without fabricating a realized product or completion marker. A valid applied policy yielding no support-authorized output rows is instead `not_produced` with cause `no_support_authorized_output_rows`; it is not failure and publishes no map, coadd member, response, covariance, or completion marker. Base-role response/covariance unavailability permitted by REQ-037–041 is a realized companion-status fact, not producer or attempt failure.

Minimum conformance evidence: Inject each failure class and the separate prospective no-support case under every product role at its exact boundary; inspect scope, lifecycle, attempt outcome, diagnostics/evidence, nonmutation, unchanged base parent, and absence of false application, realization, product identity, or completion.

Result: `not_assessed`

SCI-MAP-REQ-048 – Required observation products

Canonical clause: A complete realized logical `SCI-MAP:base_observation_map@1` bundle with at least one support-authorized scientific row shall exist before coaddition. For an applied observation attempt with no such rows, [the exact empty-support disposition](#) applies, and the attempt is not an admitted coadd member. Persisted per-observation products are the normal scientific expectation and shall be published whenever the effective v0.2 output plan designates them required; every required publication shall complete or the declared operation fails without a completion claim. A plan may instead permit a fully in-memory route. A coadd-only entry may consume persisted conforming observation bundles without rerunning upstream processing. In-memory and persisted routes shall use identical logical admission, arithmetic, role and base-parent identity, lifecycle, attempt outcome, and provenance. Optional nonpublication of a realized logical product does not imply that the product did not exist; no-support nonproduction means that no logical observation product exists. This MAP rule does not waive PTC's upstream atomic required-publication obligations.

Minimum conformance evidence: Exercise required-publication success/failure, optional in-memory nonpublication, coadd-only persisted input, and prospective no-support nonproduction for each requested role; verify identical realized base bundle content across storage routes, exclusion of a no-row attempt from coadd membership, exact failure propagation, no misleading realization/completion, and unchanged upstream PTC duties.

Result: not_assessed

SCI-MAP-REQ-049 – Immutable base parent

Canonical clause: Every stronger observation/coadd role and downstream transformation shall preserve the exact resolvable base/unfiltered MAP-bundle or base-coadd identity, ordered membership, lifecycle, and MAP-local base-product validity. A later local validity/eligibility decision may differ for its named use but shall not rewrite MAP validity, alter the original claims, sever parentage, or select a hidden subset of an existing base parent. Later response or covariance estimates are new versioned products.

Minimum conformance evidence: Create stronger roles and later locally eligible products over MAP-invalid and MAP-valid parents; verify exact membership, both validity states, parentage, and original claims remain immutable.

Result: not_assessed

SCI-MAP-REQ-050 – Complete consumer boundary

Canonical clause: No consumer shall receive a bare signal plane as a scientifically complete map. The boundary shall include exact product role, request, base parent and membership; quantity and nominal-beam identity; estimator/normalization/coefficient/profile identity; honest response class/state and uncertainty/covariance disclosure; distinct map and exposure facts; PTC additive-reference/removed-component/null-space state; exact frame/WCS, shape/indexing, all reached lifecycle stages, parentage, dependency versions, comparison-space record when applicable, and failures. Later analyses choose their own exact claim requirements and may attach honest versioned response/covariance products; MAP does not own their eligibility. SCI-NOI owns its exact realization-generation, empirical-uncertainty, conditional-scale, and standardization methods and products. Any physical- noise, total-uncertainty, covariance-completeness, significance, probability, false-rate, completeness, purity, or catalogue interpretation requires a separately approved exact method and remains unavailable by default. MAP therefore keeps Q normalization, formal conditional precision, complete conditional covariance, the NOI conditional_detector_sign_randomization_marginal_second_moment, the NOI standardized signal, and any future calibrated-significance method as distinct identities and claims.

Minimum conformance evidence: Exercise every role and claims with missing required facts, including NOI-related claims and analyses that explicitly accept limitations; verify fail-closed claim scope, exact parent/membership, and no rewrite of MAP validity.

Result: not_assessed

SCI-MAP-REQ-051 – Same fixed noise operator

Canonical clause: An admitted noise realization may be described as using the same map/coadd operator only when eligibility, projection, grouping, coefficients, normalization, exact support-authorized row selection, boundary, and atomic coadd admission are fixed or identically prescribed. Re-estimation from the randomized data defines a different named estimator.

Minimum conformance evidence: Propagate fixed deterministic vectors through signal and realization paths and compare row identities and operator coefficients, not filenames or shapes.

Result: not_assessed

SCI-MAP-REQ-052 – Method and claim separation

Canonical clause: Pointing and OOF may be registered consumers of the exact ordinary MAP gridding operator only when their inputs satisfy this operator contract. Reuse grants no mode-specific WCS/frame, source model, fitting, response interpretation, output, default, or qualification authority; an unsupplied mode profile remains unavailable. This authority shall not otherwise be applied by analogy to JINC, maximum-likelihood mapmaking, OOF residual transfer, reprojection, filtering, empirical noise, fitting, Beammap inference, feedback, Stokes Q/U, measured-R mapmaking, or other units. Independent inspection, modification, experimentation, and proposal remain unrestricted, while an altered scientific identity or official claim requires an explicit versioned successor and evidence appropriate to that claim. Algebraic contract correctness, engineering conformance, representation/response fidelity, observational performance, and production readiness shall remain separate claims.

Minimum conformance evidence: Exercise conforming and nonconforming Pointing/OOF operator requests, missing mode profiles, each prohibited method/claim promotion, and an independently changed support algorithm; verify exact operator reuse boundaries, unavailable unsupplied authority, versioned identity, and independent claim-state fields.

Result: not_assessed

6 Prediction execution matrix

Predictions are imported from the same shared authority. Engineering records attach fixtures, bounds, artifacts, and results; they do not rewrite the prediction.

SCI-MAP-PRED-001 – Uniform constant

Canonical prediction: For fixed membership, a constant admitted input with uniform positive coefficients produces that constant on every support-authorized MAP-valid output row; numerator and normalization scale together, and no unsupported scientific row is created.

Primary claim layer: Algebraic contract.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-002 – Unequally weighted constant

Canonical prediction: For fixed membership, unequal positive coefficients still preserve the same constant on every support-authorized output row, while normalization and conditional variance generally change with position; rows outside that set have no normalized scientific value.

Primary claim layer: Algebraic contract.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-003 – One-pixel hand calculation

Canonical prediction: For an authorized row with admitted z_i and $a_i > 0$, the exact triplet is $(N, Q, \hat{m}) = (\sum a_i z_i, \sum a_i, \sum a_i z_i / \sum a_i)$; an unauthorized row has no normalized scientific value.

Primary claim layer: Algebraic contract.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-004 – One-hot boundary

Canonical prediction: A finite coordinate inside exactly one half-open cell has one $G_{pi} = 1$, preserves unity over pixels, and contributes only there. A coordinate on an internal upper edge belongs to the adjacent cell; a coordinate on the outer upper extent or outside the grid contributes nowhere. Fractional splitting is not authorized.

Primary claim layer: Algebraic contract.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-005 – Delta input

Canonical prediction: For a declared fixed state, a basis-pixel sky perturbation produces the corresponding support-row-restricted column of $R_{\text{fixed}, \Theta} = A_{\text{MAP}, \Pi} H_{\text{fixed}, \Theta}$, including projection and boundary truncation. A PTC full-procedure or PTC+MAP re-resolved numerical difference exists only when its complete authorized comparison-space record supplies exact common domains and both row maps; absent that record it is unavailable or only a typed state transition/discontinuity and need not equal the fixed-state column.

Primary claim layer: Representation fidelity.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-006 – Point-source template

Canonical prediction: A declared point-source perturbation τ_b produces $R_{\text{fixed}, \Theta} \tau_b$ only for the exact fixed response state and authorized rows. PTC full-procedure and PTC+MAP re-resolved families additionally retain baseline/perturbed identities, perturbation side/sign/amplitude/location/source state/normalization, finite-difference convention, gauge/support/bases, exact common domain and row maps, state transition, availability, and cause. The unavailable whole-chain RTC-to-CAL-to-PTC-to-MAP family is not inferred. Peak, integrated, and fitted amplitudes remain distinct functionals.

Primary claim layer: Representation fidelity.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-007 – Resolved and Fourier inputs

Canonical prediction: At fixed state, resolved disks, gradients, and Fourier modes produce their explicit $R_{\text{fixed}, \Theta} \tau$ on the authorized row set; a constant-input pass does not predict preservation of these modes, and procedure-response families are not inferred from the fixed matrix.

Primary claim layer: Representation fidelity.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-008 – Variable coverage

Canonical prediction: Fixed-coefficient constant preservation can coexist with spatially varying normalization, support-row membership, covariance, and nonconstant-template response; a row that fails support is absent rather than measured zero.

Primary claim layer: Algebraic contract.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-009 – Finite boundary

Canonical prediction: At an admitted finite edge, no contribution wraps or clamps; local normalization may preserve a constant while compact-source peak or integrated response changes.

Primary claim layer: Representation fidelity.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-010 – Coefficient states

Canonical prediction: The admitted uniform family yields exact dimensionless $\gamma_i = 1$ only for an explicit selection with complete PTC publication, exact target/payload/occurrence association, MAP permission, and a requested, applicable, eligible, realized named handoff under the exact compatible source binding. Registry presence alone realizes nothing. Missing selection or handoff remains unavailable with no fallback. Structural availability is independent of numerical class; MAP alone classifies finite positive, exact zero, negative finite, non-finite, and unrepresentable. Zero is noncontributing and the other nonpositive/non-finite classes are invalid with exact cause; uniform one is not precision.

Primary claim layer: Engineering conformance.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-011 – Use-qualified exclusion and non-finite states

Canonical prediction: A candidate whose typed gate record projects to Boolean nonmembership does not contaminate an accumulator. Structural exclusion prevents payload retrieval; structurally admitted payloads are read only for safe MAP numerical classification before membership and arithmetic. Its complete named-field or other typed record, immutable reason, and cause remain preserved; no universal producer-flag action is inferred.

Primary claim layer: Engineering conformance.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-012 – Threshold selector

Canonical prediction: For each predeclared positive finite coefficient population, the selected zero-based index is exactly $\lfloor ([0.75N] + N)/2 \rfloor$; empty input uses the exact boundary convention $Q_\star = 0$ while selecting no population element and asserting no zero-sky measurement. No row gains support without its own finite positive Q_p . Small N fixtures straddle every integer-index transition. At exact $c = 0$, both relative thresholds are zero and the finite-positive requirement still excludes zero, negative, non-finite, or absent normalization. Values $0 < c \leq 1$ use inclusive comparisons in the ordinary recommended range. A $c > 1$ value is admitted only with the independently true, same-value/scope `coverage_cut_above_one_override_requested` fact and may yield empty support; the number does not authorize itself. That prospective applied case has state `applied_no_support_authorized_output_rows`, outcome `not_produced`, exact cause `no_support_authorized_output_rows`, no failure, and no realized MAP product. Negative, non-finite, missing, or unrepresentable c fails before application or required-product mutation. Every reached cut stage is retained, with all five only for successful product realization; pre-application validation checks the observation-resolved value and never requires a realized value.

Primary claim layer: Algebraic contract.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-013 – Missing response

Canonical prediction: Every `SCI-MAP:base_observation_map@1` discloses its known MAP-local realized operator response on exact signal rows. Missing upstream response yields a separately typed limited or unavailable state on complete signal membership, never a smaller subset, zero, or identity. `SCI-MAP:response_bearing_observation_map@1` is unavailable for that exact base parent only when its exact selected role request names that exact response-family object as required; absence of a different fixed-state, full-procedure, re-resolved, coadd-procedure, or whole-chain family does not block an otherwise complete named response role. Each numerical full-procedure difference requires its complete `SCI-MAP-FULL-PROCEDURE-COMPARISON-SPACE` record. Coadd-full-procedure/re-resolved and whole-chain responses remain separately unavailable unless authorized. The base bundle remains usable for claims that do not require the absent response; any later response estimate or corrected map has a new versioned identity bound to the original.

Primary claim layer: Engineering conformance.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-014 – Role-qualified atomic rejection

Canonical prediction: After exact activation makes the proposed aggregate profile evaluable, a required companion incompatibility rejects the complete observation before coadd mutation at the selected role scope. Under `SCI-MAP:base_coadd@1`, unavailable or basis-incompatible response does not change signal membership; it yields unavailable coadd response with exact cause and no hidden subset or zero companion. Under `SCI-MAP:response_bearing_coadd@1`, a mixed or incompatible response-family identity, missing fixed-state condition, or absent exact composition theorem blocks that role for an existing base parent. A required covariance incompatibility analogously blocks `SCI-MAP:covariance_qualified_coadd@1`. Any separately requested admission may reject the observation only before constructing its own explicitly identified base/signal parent. Other required identity, unit, WCS, shape, policy, and parent incompatibilities remain atomic at their declared role scope. The current candidate has no active profile or realized object decision.

Primary claim layer: Engineering conformance.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-015 – One-observation coadd

Canonical prediction: After exact profile activation and a realized eligible object decision, a `SCI-MAP:base_coadd@1` containing one admitted base observation under the uniform dimensionless coefficient family reproduces its signal on corresponding centered-integer authorized rows. If the valid applied coadd support policy authorizes no rows, outcome is instead `not_produced` with exact no-support cause and no zero coadd. Conforming in-memory and persisted bundles give the same admission, identity, and arithmetic. Unavailable or basis-incompatible upstream response leaves nonempty signal membership unchanged and makes coadd response unavailable with exact cause. `SCI-MAP:response_bearing_coadd@1` reproduces response only when it binds one exact family R , every $R_o^{(R)}$ and all fixed-state/composition conditions are compatible, and $R_{\text{coadd,out}}^{(R)} = B_{\text{out}} T_{\text{obs}}^{(R)}$ is authorized.

Primary claim layer: Algebraic contract.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-016 – Uniform observation coefficients

Canonical prediction: On each authorized coadd row, two compatible observations produce their equal-observation arithmetic mean and count-like normalization under `SCI-MAP:uniform_observation_coadd_coefficient@1`. Each requested coadd role binds its own exact parent, lifecycle, attempt outcome, and ordered membership; stronger companions do not change the arithmetic membership of their exact base parent. A response-bearing coadd additionally binds one exact compatible response family and the fixed coadd state/composition authority; a finite difference without an exact composition theorem is not propagated as a Jacobian. Coadd-full-procedure/re-resolved and whole-chain response remain unavailable. The normalization is not precision.

Primary claim layer: Algebraic contract.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-017 – Missing, invalid, or no-support observation

Canonical prediction: An unsupported observation row contributes at no coadd row; an observation rejected by the exact selected role contributes nowhere to that separately identified base/signal parent and does not increment its count. An applied observation attempt with no support-authorized output rows has outcome `not_produced`, is not an admitted coadd member, and has no signal, response, covariance, exposure, count, validity, or completion product. A companion unavailable but permitted by a realized base role does not remove the observation and is not attempt failure. No case creates a zero-valued scientific or companion substitute.

Primary claim layer: Engineering conformance.

Fixture / bound / result: `not_assessed`; preregistration pending

SCI-MAP-PRED-018 – Centered integer placement

Canonical prediction: A canonical target grid requested before construction yields observation maps already bound to that grid. Other compatible shape differences that are nonnegative and even produce the exact half-difference row/column offset; odd, negative, fractional, or WCS-inconsistent relations fail with no mutation. The FITS representation comparison evaluates every preregistered addressable pixel center and outer-boundary cell vertex, including all corners, with the exact spherical metric and invalid-point rule. A later crop is a new row-selected product with its own identity and lineage and cannot relabel its parent as compatible.

Primary claim layer: Engineering conformance.

Fixture / bound / result: `not_assessed`; preregistration pending

SCI-MAP-PRED-019 – Cross-observation covariance

Canonical prediction: For two maps with variances σ_1^2, σ_2^2 and covariance c , a complete `SCI-MAP:covariance_qualified_coadd@1` over the exact base-coadd membership includes c through Σ^{obs} . If a required block is missing, that stronger role is unavailable for the parent; the base role retains a partial, symbolic, summarized, lineage-resolvable, or unavailable state. No subset is formed, the block is never zero or independence, and equal-observation normalization is not precision.

Primary claim layer: Algebraic contract.

Fixture / bound / result: `not_assessed`; preregistration pending

SCI-MAP-PRED-020 – Fixed noise operator

Canonical prediction: A deterministic admitted realization propagated with fixed state yields the same support-row identities and coefficient-by-coefficient $A_{\text{MAP},\Pi} \equiv A_{\text{out}}$ and B_{out} as the signal path; recomputed state exposes a named difference.

Primary claim layer: Engineering conformance.

Fixture / bound / result: `not_assessed`; preregistration pending

SCI-MAP-PRED-021 – Sequential and parallel reduction

Canonical prediction: Integer facts, identities, masks, support, and cardinality agree exactly. Floating sums agree with a preregistered operation-count/conditioning bound that is frozen before results are viewed and is not loosened after failure.

Primary claim layer: Engineering conformance.

Fixture / bound / result: `not_assessed`; preregistration pending

SCI-MAP-PRED-022 – No false significance

Canonical prediction: A deliberate attempt to label signal times square root of merely formal weight as empirical significance is rejected or remains unmistakably labeled formal standardized signal.

Primary claim layer: Engineering conformance.

Fixture / bound / result: `not_assessed`; preregistration pending

SCI-MAP-PRED-023 – No validity promotion

Canonical prediction: A downstream product may have its own use-local validity or eligibility but cannot change MAP-local base-product validity or sever the base/unfiltered MAP parent.

Primary claim layer: Engineering conformance.

Fixture / bound / result: `not_assessed`; preregistration pending

SCI-MAP-PRED-024 – No ordinary predicate for JINC

Canonical prediction: A deliberate attempt to apply positive-coefficient ordinary admission, support, or complete-bundle availability to JINC is rejected as outside SCI-MAP v0.2.

Primary claim layer: Engineering conformance.

Fixture / bound / result: not_assessed; preregistration pending

SCI-MAP-PRED-025 – Quantity-specific zero, index, and no-support outcomes

Canonical prediction: A finite exact-zero signal numerator $N_p = 0$ with finite positive normalization $Q_p > 0$, valid support, and every other required gate passing is admissible and gives the valid normalized scientific value $\hat{m}_p = 0$. An in-bounds zero-based index and a zero centered-integer placement offset are valid when their exact domain and embedding rules pass. A row with $Q_p = 0$ is outside finite-positive normalization support: division is unauthorized and no supported scientific row is formed. This normalization-specific result is not a generic rule that zero is invalid. A zero coefficient remains noncontributing; **coverage_cut** = 0 retains its no-relative-cut meaning; and support-population count $N = 0$ retains $Q_\star = 0$ while selecting no population element and authorizing no row by itself. Non-finite, overflowed, out-of-domain, and unrepresentable quantities follow their exact declared failure scopes before affected mutation or completion. Separately, a valid applied policy whose support-authorized output-row set is empty produces no product with exact **not_produced** outcome and cause **no_support_authorized_output_rows**; it is not a failure and creates no fabricated zero substitute.

Primary claim layer: Engineering conformance.

Fixture / bound / result: not_assessed; preregistration pending

7 Mechanical completeness checks

Before a conformance verdict, automated checks shall establish:

- exactly one declaration of each stable requirement and prediction ID in the shared authority, with no gaps or duplicates;
- every requirement appears in this specification and in `CROSSWALK.md`; every prediction is linked to at least one requirement;
- lifecycle, attempt outcome, empty-support disposition, six profile-status axes, exact response-family coadd construction, the stable comparison-space record, NOI terminology, and document-authority roles agree across the core, three views, records, profiles, bindings, and prospective templates;
- each rendered PDF identifies contract v0.2 and its current document revision separately;
- no unresolved LaTeX reference, citation, overfull box, clipped glyph, broken table, or unreadable page remains;
- every required artifact reopens successfully, has the expected page count, and renders every page through an independent PDF renderer; and
- the owner-decision ledger and author-draft decision record contain no silent resolution of an open scientific question.

The conformance verdict uses the same exact five-value requirement-result vocabulary. It is **pass** only when every applicable requirement is **pass** on the exact candidate. Any contradictory evaluated requirement makes it **fail**; otherwise any missing named prerequisite makes it **blocked**; otherwise any applicable unevaluated requirement makes it **not_assessed**. **not_applicable** is permitted only for a requirement outside the exact candidate's declared role with an owner-approved rationale. Evidence-artifact **incomplete** is never a requirement result. This document records every placeholder as **not_assessed** and asserts no verdict.

Required handoff contents

An engineering handoff contains the candidate and authority identities; requirement result table using exactly **pass**, **fail**, **blocked**, **not_applicable**, and **not_assessed**; a separate artifact-realization field using exactly **complete**, **incomplete**, **failed**, and **not_produced**; a separate MAP attempt outcome using **succeeded**, **failed**, and **not_produced**; product-realization identity only for actual products; prediction result table; exact commands and environment; independent expected-value artifacts; build/test/log outputs; representation and observational evidence where claimed; failure-injection snapshots; skipped/blocked items; deviations and owner dispositions; and an explicit statement for each claim layer. The handoff shall not collapse “conforms algebraically,” “represents faithfully,” “performs on sky,” and “ready for production” into one status word.